

Tab 2

Task1_Documentation.pdf

INTERN & PROJECT DETAILS

COMPANY NAME. : CODTECH IT SOLUTIONS

INTERN NAME. : MALLAVARAPU LAVANYA

INTERN ID. : CITS1289

DOMAIN. : VLSI DESIGN

BATCH DURATION. : MAY-JUNE 2026

TASK TITLE : 7-SEGMENT DISPLAY DRIVER

PROJECT OVERVIEW

This project implements a 4-bit Binary Coded Decimal (BCD) to 7-segment display decoder in Verilog. It converts a 4-bit digital input (representing values from 0 to 9) into the corresponding 7-bit control signal needed to drive a standard 7-segment display using active-high logic.

Design Specifications

- **Hardware Description Language:** Verilog HDL
- **Target Domain:** VLSI Design / Digital Systems
- **Logic Type:** Active-High (1 turns segment ON, 0 turns segment OFF)
- **Output Format:** 7-bit bus ordered as [g f e d c b a]

Segment Mapping Truth Table

Input (Dec)	BCD Input [3:0]	Output [g f e d c b a]	Displayed Character
-------------	--------------------	---------------------------	------------------------

0	0000	0111111	0
1	0001	0000110	1
2	0010	1011011	2
3	0011	1001111	3
4	0100	1100110	4
5	0101	1101101	5
6	0110	1111101	6
7	0111	0000111	7
8	1000	1111111	8
9	1001	1101111	9
INVALID	1010 to 1111	0000000	Blank

SOURCE CODE

//

=====

=====

```
// COMPANY           : CODTECH IT SOLUTIONS
// INTERN. NAME      : MALLAVARAPU LAVANYA
// INTERN ID.        : CITS1289
// DOMAIN.           : VLSI
// TASK TITLE.       : MAY-JUNE 2026
//.
=====
=====
```

VERILOG DESIGN CODE

```
`timescale 1ns / 1ps

module seven_seg_driver (
    input wire [3:0] bcd_in,    // 4-bit BCD input (0 to 9)
    output reg [6:0] seg_out    // 7-bit output for segments
    (g f e d c b a)
);

    // Combinational logic block for BCD to 7-Segment
    conversion
    always @(*) begin
        case (bcd_in)
            // Logic: 1 = Segment ON, 0 = Segment OFF
            // Mapping: seg_out = {g, f, e, d, c, b, a}

```

```

    4'b0000: seg_out = 7'b0111111; // Displays 0
    4'b0001: seg_out = 7'b0000110; // Displays 1
    4'b0010: seg_out = 7'b1011011; // Displays 2
    4'b0011: seg_out = 7'b1001111; // Displays 3
    4'b0100: seg_out = 7'b1100110; // Displays 4
    4'b0101: seg_out = 7'b1101101; // Displays 5
    4'b0110: seg_out = 7'b1111101; // Displays 6
    4'b0111: seg_out = 7'b0000111; // Displays 7
    4'b1000: seg_out = 7'b1111111; // Displays 8
    4'b1001: seg_out = 7'b1101111; // Displays 9
    default: seg_out = 7'b0000000; // Turns all
segments OFF for invalid BCD
    endcase
end

endmodule

```

TEST BENCH CODE

```
=====
```



```

=====
// COMPANY.      : CODE IT SOLUTIONS
//. INTERN NAME. : MALLAVARAPU LAVANYA
//. INTERN ID.   : CITS1289
//. DOMAIN.      : VLSI DESIGN
//. TASK TITLE.  : 7-SEGMENT DISPLAY DRIVER
//
=====
=====

```

TEST BENCH CODE

```

`timescale 1ns / 1ps

module tb_seven_seg_driver;

    // Inputs to the Design Under Test (DUT)
    reg [3:0] bcd_in;

    // Outputs from the DUT
    wire [6:0] seg_out;

    // Instantiate the Unit Under Test (UUT)
    seven_seg_driver uut (
        .bcd_in(bcd_in),
        .seg_out(seg_out)
    );

```

```

);

// Stimulus generation block
initial begin
    // Setup value dumping for waveform generation
    $dumpfile("dump.vcd");
    $dumpvars(0, tb_seven_seg_driver);

    // Monitor system changes on the console terminal
    $monitor("Time = %0t ns | BCD Input = %b (%d) |
7-Seg Output (g-f-e-d-c-b-a) = %b",
            $time, bcd_in, bcd_in, seg_out);

    // Test Case 0: Initialize
    bcd_in = 4'b0000; #10;

    // Test Cases 1 to 9: Loop through all valid BCD
values
    bcd_in = 4'b0001; #10;
    bcd_in = 4'b0010; #10;
    bcd_in = 4'b0011; #10;
    bcd_in = 4'b0100; #10;
    bcd_in = 4'b0101; #10;
    bcd_in = 4'b0110; #10;
    bcd_in = 4'b0111; #10;
    bcd_in = 4'b1000; #10;
    bcd_in = 4'b1001; #10;

```

```
// Test Case 10: Verify invalid BCD input handling
(Default Case)
bcd_in = 4'b1111; #10;

// Terminate the simulation run cleanly
$finish;
end

endmodule
```

Simulation & Verification Results

Expected Console Output

When you compile and run this code on an EDA simulator (like EDA Playground, ModelSim, or Vivado), the console output will look exactly like this

10:5902:49

Edit code - EDA Playground
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EDA playground

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Languages & Libraries

Testbench + Design
SystemVerilog/Verilog

UVM / OVM
None

Other Libraries
None
OVL
SVUnit

☐ Enable TL-Verilog
☐ Enable Easier UVM
☐ Enable VUnit

Tools & Simulators
Icarus Verilog 12.0

Compile Options
-Wall -g2012

Run Options
☐ Use run.bash shell script
☒ Open EPWave after run
☐ Show output text file
☐ Show output SVG file
☐ Download files after run

Examples
using EDA Playground
VHDL
Verilog/SystemVerilog
UVM
EasierUVM
SVAUnit
VUnit (Verilog/SV)
VUnit (VHDL)
TL-Verilog
e + Verilog
Python + Verilog Python + VHDL
Python Only
C++/SystemC

testbench.sv

design.sv

1 // Company: CODETECH IT SOLUTIONS
2 // Domain: VLSI
3 // Intern Name: Mallavarapu Lavanya
4 // Intern ID: CITS1289
5 // Task Title: Testbench for 7-Segment Display Driver
6
7 timescale 1ns / 1ps
8 module seven_seg_driver (
9 input wire [3:0] bcd_in, // 4-bit BCD input (0 to 9)
10 output reg [6:0] seg_out, // 7-bit output for segments (g f e d c b a)
11
12 // Behavioral block that updates whenever the BCD input changes
13 always @(*) begin
14 case (bcd_in)
15 // Active-high logic (1 turns the segment ON, 0 turns it OFF)
16 // Segment
17 order: g-f-e-d-c-b-a
18 seven_seg_driver uut (
19 .bcd_in(bcd_in),
20 .seg_out(seg_out)
21);
22 // Stimulus process to test all BCD inputs
23 initial begin
24 \$dumpfile("dump.vcd");
25 \$dumpvars(0,
26 tb_seven_seg_driver);
27 // Monitor the changes in input and output lines
28 \$monitor("Time = %0t | BCD Input = %b (%d) | 7-Seg Output (g-f-e-d-c-b-a) = %b", \$time, bcd_in, bcd_in, seg_out);
29
30 // Initialize
31 input bcd_in = 4'b0000;
32
33 // Loop through all valid BCD values (0 to 9)
34 #10; bcd_in = 4'b0001;
35 #10; bcd_in = 4'b0010;
36 #10; bcd_in = 4'b0011;
37 #10; bcd_in = 4'b0100;
38 #10; bcd_in = 4'b0101;
39 #10; bcd_in = 4'b0110;
40 #10; bcd_in = 4'b0111;
41 #10; bcd_in = 4'b1000;
42 #10; bcd_in = 4'b1001;

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14 always @(*) begin
15 case (bcd_in)
16 // Active-high logic (1 turns the segment ON, 0 turns it OFF)
17 // Segment
18 order: g-f-e-d-c-b-a
19 seg_out = 7'b0111111; // Displays 0
20 seg_out = 7'b0000110; // Displays 1
21 seg_out = 7'b0100111; // Displays 3
22 seg_out = 7'b1001101; // Displays 4
23 seg_out = 7'b1011011; // Displays 5
24 seg_out = 4'b0110; // Displays 6
25 seg_out = 7'b1111111; // Displays 7
26 seg_out = 4'b0000; // Displays 8
27 seg_out = 7'b1101111; // Displays 9
28 default: seg_out = 7'b0000000; // Default case turns all segments OFF
29 endcase
30 end
31 endmodule

Log

Share

2026-06-05 05:28:26 UTC | verilog -Wall -g2012 design.sv testbench.sv
VCD info: dumpfile dump.vcd opened for output.
Time = 0 | BCD Input = 0000 (0) | 7-Seg Output (g-f-e-d-c-b-a) = 0111111
Time = 10000 | BCD Input = 0001 (1) | 7-Seg Output (g-f-e-d-c-b-a) = 0000
Time = 20000 | BCD Input = 0010 (2) | 7-Seg Output (g-f-e-d-c-b-a) = 1011
Time = 30000 | BCD Input = 0011 (3) | 7-Seg Output (g-f-e-d-c-b-a) = 1001
Time = 40000 | BCD Input = 0100 (4) | 7-Seg Output (g-f-e-d-c-b-a) = 1100
Time = 50000 | BCD Input = 0101 (5) | 7-Seg Output (g-f-e-d-c-b-a) = 1101
Time = 60000 | BCD Input = 0110 (6) | 7-Seg Output (g-f-e-d-c-b-a) = 1111
Time = 70000 | BCD Input = 0111 (7) | 7-Seg Output (g-f-e-d-c-b-a) = 0000
Time = 80000 | BCD Input = 1000 (8) | 7-Seg Output (g-f-e-d-c-b-a) = 1111
Time = 90000 | BCD Input = 1001 (9) | 7-Seg Output (g-f-e-d-c-b-a) = 1101
Time = 100000 | BCD Input = 1111 (15) | 7-Seg Output (g-f-e-d-c-b-a) = 000
testbench.sv:48: \$finish called at 110000 (1ps)
Simulation to fail0:

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2026-06-05 05:28:27 UTC | Opening EPWave... EPWave

10:5903:10

Edit code - EDA Playground
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EDA playground

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VUnit (VHDL)
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21 seg_out = 7'b1001101; // Displays 4
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23 seg_out = 4'b0110; // Displays 6
24 seg_out = 7'b1111111; // Displays 7
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cr CODEGENOS
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27 default: seg_out = 7'b0000000; // Default case turns all segments OFF
28 endcase
29 end
30 endmodule

Log

Share

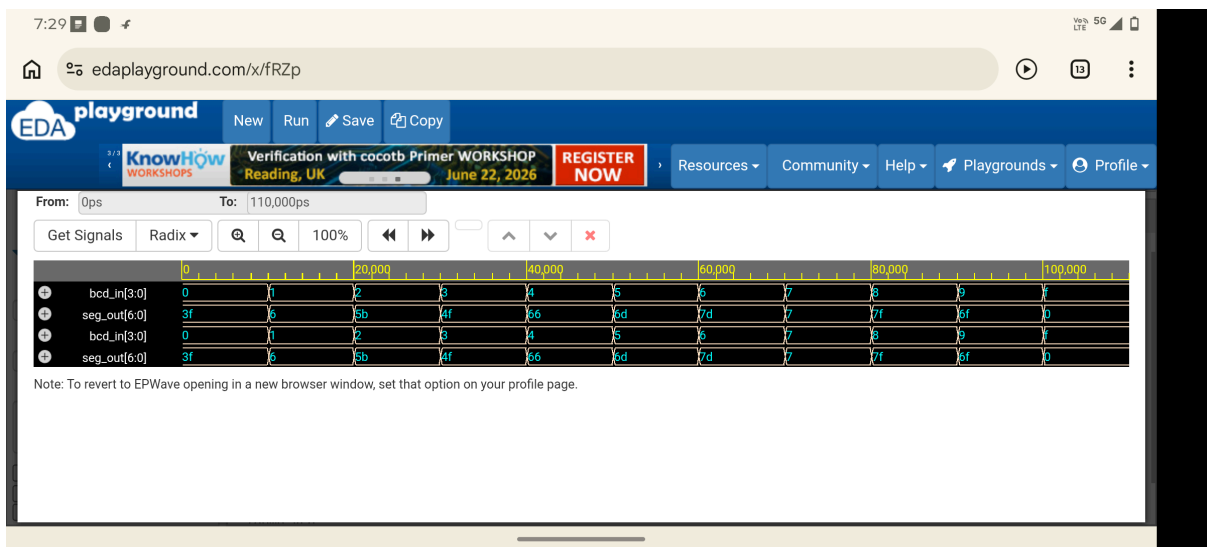
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Time = 70000 | BCD Input = 0111 (7) | 7-Seg Output (g-f-e-d-c-b-a) = 0000
Time = 80000 | BCD Input = 1000 (8) | 7-Seg Output (g-f-e-d-c-b-a) = 1111
Time = 90000 | BCD Input = 1001 (9) | 7-Seg Output (g-f-e-d-c-b-a) = 1101
Time = 100000 | BCD Input = 1111 (15) | 7-Seg Output (g-f-e-d-c-b-a) = 000
testbench.sv:48: \$finish called at 110000 (1ps)
Simulation completed

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2026-06-05 05:28:27 UTC | Opening EPWave... EPWave

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TEXT :



CONCLUSION:

The design successfully translates BCD values into appropriate signals for a 7-segment display. The testbench verifies all operations, including proper blanking when encountering invalid inputs outside the 0–9 range. If you need help generating a formal project report or need to convert this code to an active-low configuration for a common anode display, let me know!

Tab 3

